Kartikey Sondhi

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Forward Deployed Engineer (FDE) – LLM Applications, Guardrails & Evaluation

Full-stack AI engineer with about two years of experience taking LLM products from prototype to production at Deloitte: guardrails, evaluation infrastructure, and a chatbot over 25,000 documents, with measurable gains in accuracy (22% to 90% pass rate), latency (∼20x faster), and throughput (12–14 hours to about 2). Also built and shipped Usovo, a local-first AI audio studio, solo and end to end. First author of a VEHITS 2024 paper. Creating Interro.

Experience

Data Scientist 2, Deloitte *(Product Engineering team)* Jul 2025 – Present
M&A due-diligence platform: web scanning, structured findings, risk flags, and a chatbot over internal documents.

- **Cut query latency by ∼20x with pre-LLM fast path**. Designed a repository-layer direct matcher that bypasses the 3-step LLM pipeline for bare company names and common phrasing, reducing response time from **14-16 sec to 0.3-1sec** while, reducing handoff to LLMs only when queries are complex in nature.
- Raised end to end pass rate for direct matcher by **22% -> 92%** by fixing systematic LLM SQL-generation failure classes, by introducing **% wildcard matching**, **acronymalias resolution** and **prefix anchoring** to fix acronym overmatching eliminating all pipeline errors across 100 diverse evals with **zero regression**.
- Built an automated eval and failure-diagnosis infrastructure. An offline miss classifier for empty-results, pipeline-errors, wrong-set with per category and per path pass rates, plus a SQL query replay probe that distinguishes LLM output corruption from data issues, **cutting throttling-induced noise by 76%** for more than one project.
- Rebuilt an early prototype (unstructured code, basic AI logic) into a production pipeline for scraping, classification, anomaly detection, and flagging, on a six-engineer team that later shrank to two (a senior developer and me).
- Built an extraction and cleaning pipeline for about **25,000 documents (about 40,000 characters each)** creating chunked, labeled and processed chunks for retrieval.
- Redesigned due diligence report creation pipeline around async and parallel execution, cutting runs from **12–14 hours to about 2 hours (about 85%)**.
- Improved retrieval and classification to **96% precision**, with **+18% answer accuracy** and **94% semantic query re-framing accuracy**; raised recall and built the datasets behind these gains.
- Added output guardrails to user facing chatbot for faulty, incomplete, gibberish, unstructured, and hallucinated responses, **eliminating up to 98%** faulty outputs.
- Improved retrieval accuracy and contextual reasoning by implementing a Graph RAG system in Neo4j with caption-metadata-enriched knowledge graphs.
- Designed and presented learning modules, to over 200 people, teaching proper and efficient use of tokens when dealing with coding agents, with topics like integrating **Graphify** and using skills. Other topics include creating **pytest automations**, **dataset creation techniques** and using **tools and mcp servers**.

Data Science Intern, Deloitte Jan 2025 – July 2025

- Integrated corrective RAG, improving accuracy by 18% for queries related to Indian accounting standards.
- Designed agentic workflows for audit-related automation, involving integrating OCR + Vision model pipelines that improved document data extraction **accuracy by 21%**.

Projects

Interro – Voice First Interrogation Detective Game *(In progress)*

- I built a C++17 engine for a voice-driven detective game, where AI suspects react to how the player speaks, not what is typed or selected. **Parallel threads** capture voice, **Whisper.cpp does speech-to-text** and **WavLM extracts paralinguistics (378 MB ONNX model, 6 acoustic signals)**. Those signals feed an integrator emotion model these conditions, the LLM's replies, and it reached 5/5 stable start/stop cycles, **1 s server startup, 1–5 ms API latency and a 372 ms median tick (against a 400 ms target)** in the Python prototype.
- Ran a **378 MB WavLM-base-plus model on ONNX Runtime** to extract 6 signals: frame-norm energy, cosine-distance frame delta, pitch-slope trajectory, norm variance, pause rate and speech rate. A raw-PCM fallback engine provides the same signal set. The features follow **eGeMAPS and cognitive-load research**.
- Built a "dopamine" hint system. It scores each disclosure condition by distance to threshold into **cold, warm and hot zones (>0.40, 0.15–0.40, <0.15)**, with probabilistic tells of 60% (warm) and 80% (hot) injected into the LLM prompt, plus an anti-frustration scaffold.

- Built an STT energy gate that cuts idle GPU load by skipping Whisper (Metal) inference on silent 2 s intervals (RMS < 0.003), and **filtered hallucinated or non-speech transcripts** (no-speech prob > 0.95, noise-token stripping) before they reached the LLM.

Usovo – Local-First AI Audio Studio *(solo)* usovo.in

- Built Usovo AI end to end as a solo developer: a privacy-first macOS (Apple Silicon) text-to-speech desktop app that runs models on-device, with Studio voice generation, voice cloning, a system-wide read-aloud hotkey, a Library for generations, and chaptered multi-voice audiobooks from PDF/EPUB/DOCX/TXT. It has an Electron shell, a 37-route FastAPI/MLX gateway, and a Supabase and PayPal subscription backend.
- Hotkey generation is TTS at one shortcut, in which read-aloud to first audible audio of about **0.4–0.8 s** on short selections **(median 2.4 s across all 17 logged presses, max 3.3 s, up to 4,874-character selections)** by pre-warming the model at launch **(median 6.1 s)** and synthesizing a 60-character lead-in segment first.
- Built **per-model text normalization and chunking**: abbreviation-protected sentence splitting, repeat-run collapsing, a 400-character run-on hard cut, and a 60-character lead-in split for first-audio latency. It also covers heading detection (PDF outline, EPUB TOC, regex cascade, font clustering, LLM refinement) and rules-first speaker attribution, with a 60-fixture stress set across 36 edge-case categories, for audiobook creation.
- Built voice cloning with guided recording and consent UI; root-caused unstable clones to a reference-audio/transcript mismatch and cut worst-case generation from **345s to under 7s** with input-length-aware token limits.

Lifelong Dynamic Timed A* (LTA*) *(first author)* Paper

- Extended Lifelong Planning A* with arrival-time-aware edge costs from predicted speeds, for fastest-path retrieval and real-time replanning in congested road networks. Published at VEHITS 2024 (Angers, France).
- In simulation, cut travel time by **29–48%** versus A* across 500×500 and 1000×1000 grids and a Washington urban map, at **0.6–2.1%** longer routes on the grids.

Skills

Languages & Frontend: Python, JavaScript, SQL, React, Electron
Backend & Data: FastAPI, REST APIs, Redis, async/parallel pipelines, PostgreSQL, Docker, Git
AI: LLM APIs, LLM agents, RAG, retrieval and evaluation pipelines, output guardrails, prompt-injection detection, sentence embeddings, Apple MLX
Agentic tooling: Claude Code and other coding agents, MCP, tool calling, agent integration, scheduled triggers, automation bots

Education

The LNM Institute of Information Technology, B.Tech and M.Tech, Computer Science Sept 2020 – May 2025